

Ilakiya Sulur Ravi Sumathi

MSc Data Science, Leiden University

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Technical Skills

Machine Learning: PyTorch, scikit-learn, pandas, NumPy
Reinforcement Learning: REINFORCE, DQN, A2C / actor-critic (Gymnasium)
NLP: Transformers (RoBERTa), TF-IDF, cross-validation
Data & Graph: SciPy sparse matrices, MinHash/LSH, NetworkX, K-Means
Programming: Python, Java, SQL, MongoDB
Tools: GitHub, Jupyter, L^AT_EX, Power BI

Research

- Identification of Contrails using Machine Learning, *Industrial Engineering Journal*, Vol. 53(6), 2024.
- Presented the contrail-detection research at the International Conference on Computer Communication and Informatics (ICCCI), 2024.

Awards & Leadership

Guinness World Record - participant in the SAWAIT.AI Learnathon, which set the record for most participants in a 24 hour technical challenge (2024).

Completed a **1000 km cycling rally** promoting green cover and environmental awareness; recognised by the Assistant Chief Minister of Tamil Nadu (2023).

Chief Organising Member, ICCCI international conference - coordinated author communications and sessions over six months.

Profile

MSc Computer Science (Data Science) student at Leiden University, seeking a Master's thesis internship in the Netherlands. I work across reinforcement learning, sequential modelling and applied machine learning. I'm especially motivated by AI for the energy transition and climate. I learn quickly, care about getting evaluation right and enjoy turning messy problems into clear, testable models.

Education

MSc Computer Science - Data Science (2025 - present)
Leiden University (LIACS)
BE Computer Science (2020 - 2024)
Sri Shakthi Institute of Engineering and Technology

Projects

Reinforcement Learning for Control

Policy-gradient, actor-critic & value-based methods

- Implemented REINFORCE and Soft Actor-Critic (SAC) and benchmarked them against A2C and actor-critic baselines on CartPole-v1, comparing training stability and sample efficiency across 5 random seeds.
- Analysed the instability of naive deep Q-learning (DQN) through ablation experiments isolating the role of experience replay and target networks.

Sequential Recommendation with SASRec

Self-attention model, MovieLens 1M

- Built and evaluated a self-attention sequential recommendation model (SASRec) in PyTorch on the MovieLens 1M dataset.
- Identified and fixed a critical sampled vs full ranking evaluation bug that had inflated reported metrics.

Contrail Detection & Route Optimization

Satellite ML - published

- Built a contrail detection pipeline using CNN-based segmentation on satellite imagery with atmospheric data preprocessing.
- Used KNN and atmospheric features to predict contrail likelihood and proposed contrail-aware route alternatives to reduce climate impact.

Experience

Software Engineering Intern

Jun 2024 - Dec 2024

Teplar Solutions

- Delivered feature work, debugging and code reviews in a production Java codebase.
- Improved reliability by adding tests across common failure cases and writing reproducible handover documentation.